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Day 1 and 2 Questions

Note: Do Not depend entirely on this and study from the official slides

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Question 1

Traditional programming relies on writing explicit rules to process data. How does the Machine Learning approach differ in terms of its "inputs" and "outputs" during the training phase?

- A. It takes Rules and Data as input to produce Answers as output.
- B. It takes Data and random Noise as input to produce Answers.
- C. It takes Data and Answers as input to produce Rules as output.
- D. It takes Rules and Answers as input to produce Data as output.

Correct Answer: C

Question 2

You are training a model and notice that your Training Loss is very low, but your Validation Loss is high and increasing. What is the most likely diagnosis?

- A. Underfitting (High Bias) where the model is too simple.
- B. Overfitting (High Variance) where the model memorizes noise.
- C. The learning rate is set too low for convergence.
- D. Good Fit (Low Variance) with optimal generalization.

Correct Answer: B

Question 3

Why is Stratified K-Fold splitting often preferred over a simple random split, especially for medical datasets with rare conditions?

- A. It significantly speeds up the training process by automatically identifying and removing statistical outliers.
- B. It ensures that the model processes every single data point exactly one time during the validation phase.
- C. It automatically transforms the target variables from discrete categorical labels into continuous numerical values.
- D. It guarantees that the training and test sets have the same class ratios (e.g., preserving the 1% sick rate).

Correct Answer: D

Question 4

In the context of optimization, what is the specific role of the "Gradient" in Gradient Descent?

- A. It points in the direction of the steepest increase in loss, so we move opposite to it.
- B. It is responsible for randomly initializing the starting values of the model's weights and biases.
- C. It measures the absolute error between the model's prediction and the actual target value.
- D. It mathematically modifies the loss function to ensure it creates a strictly convex landscape.

Correct Answer: A

Question 5

You are building a cancer detection system. You want to ensure you catch every single sick patient, even if it means accidentally flagging some healthy people for further testing. Which metric should you optimize?

- A. Precision
- B. Accuracy
- C. Recall
- D. MSE

Correct Answer: C

Question 6

Why is Accuracy often a misleading metric for imbalanced datasets (e.g., 99% healthy, 1% sick)?

- A. It creates mathematical errors because it is fundamentally unable to process negative integer values in the input.
- B. It cannot be used for optimization because the function is non-differentiable at the decision boundary.
- C. It requires all input features to be strictly normalized between the values of 0 and 1 to function correctly.
- D. A "lazy" model that always predicts the majority class will get a very high score (99%) while failing the actual task.

Correct Answer: D

Question 7

Which of the following best describes the difference between Parameters and Hyperparameters?

- A. They are essentially interchangeable synonyms used to describe the exact same configuration variables.
- B. Parameters are learned from data; Hyperparameters are set before training.
- C. Parameters are exclusively used for Classification tasks, while Hyperparameters are reserved for Regression.
- D. Parameters are manually configured before training begins, whereas Hyperparameters are learned directly from data.

Correct Answer: B

Question 8

According to the lectures, when we have limited data, what problem does this create for complex models?

- A. They utilize built-in mechanisms to automatically perform feature selection and remove irrelevant data.
- B. They are computationally efficient and train significantly faster than simpler linear models.
- C. They memorize noise instead of learning the true signal.
- D. They lack the capacity to identify any meaningful patterns or relationships within the dataset at all.

Correct Answer: C

Question 9

When evaluating a regression model, you notice the dataset has many extreme outliers. Which metric would be less sensitive to these outliers?

- A. Mean Absolute Error (MAE)
- B. Mean Squared Error (MSE)
- C. R-Squared Score
- D. Root Mean Squared Error (RMSE)

Correct Answer: A

Question 10

What is the primary purpose of the F1-Score, and why is the Harmonic Mean used instead of a simple average?

- A. To accurately measure the Euclidean distance between the predicted outputs and the actual continuous targets.
- B. To provide a smooth derivative that allows the optimizer to calculate gradients more computationally efficiently.
- C. To mathematically transform a continuous regression problem into a discrete binary classification problem.
- D. To balance Precision and Recall, forcing the model to be good at both.

Correct Answer: D

Question 11

Why do we square the residuals $(y - \hat{y})^2$ in Mean Squared Error (MSE)?

- A. To fundamentally change the nature of the problem from a regression task into a classification task.
- B. To ensure that the resulting error value is always normalized to be less than 1 for stability.
- C. To prevent positive and negative errors from canceling out and to heavily penalize large errors.
- D. To intentionally create a non-differentiable function that assists in the convergence of specific optimizers.

Correct Answer: C

Question 12

In Logistic Regression, what is the specific purpose of the Sigmoid function?

- A. To calculate the precise Euclidean distance between data points in the high-dimensional feature space.
- B. To map any real-valued number into a probability range of $(0, 1)$.
- C. To directly minimize the weights of the model using a closed-form algebraic solution.
- D. To explicitly construct a linear decision boundary that physically separates the different classes.

Correct Answer: B

Question 13

You are training a classification model and your Log Loss is exploding (becoming very large). What does this indicate about your model's predictions?

- A. The model is making high-confidence predictions that are consistently correct across samples.
- B. The model is highly uncertain, outputting probabilities close to 0.5 for most inputs.
- C. The model has successfully converged and reached the global minimum of the loss function.
- D. The model is confidently wrong (predicting high probability for the wrong class).

Correct Answer: D

Question 14

What is the key distinguishing characteristic of Lasso Regression (L1) that makes it particularly useful for feature selection?

- A. It shrinks weights to exactly zero, effectively removing features.
- B. It utilizes a closed-form algebraic solution that makes it significantly faster to compute than Ridge.
- C. It applies a penalty that shrinks all weights evenly toward zero but ensures no weight ever actually reaches zero.
- D. It constrains the weights within a circular region, which mathematically prevents the coefficients from becoming sparse.

Correct Answer: A

Question 15

Why is k-NN called a "Lazy Learner"?

- A. It automatically selects only the most important features during the training phase to speed up prediction.
- B. It utilizes a gradient descent process that converges very slowly compared to other algorithms.
- C. It memorizes all training data and only computes distances at prediction time (no training phase).
- D. It halts the training process prematurely to prevent overfitting and save computational resources.

Correct Answer: C

Question 16

In k-NN, what is the main risk if you set $k = 1$?

- A. The model will automatically detect and eliminate features that are statistically irrelevant.
- B. The model will lose its non-linear capabilities and exhibit linear behavior identical to logistic regression.
- C. The decision boundary will become excessively smooth, resulting in significant underfitting of the data.
- D. The model will overfit due to high sensitivity to noise in individual points.

Correct Answer: D

Question 17

What problem does the "Curse of Dimensionality" describe, and why does it hurt model performance?

- A. Data becomes extremely sparse in high dimensions, and distances lose their meaning.
- B. Data points become tightly clustered together, which makes separating classes significantly easier.
- C. The time required to train the model decreases because high dimensions provide more separation paths.
- D. The algorithm automatically performs feature selection, significantly reducing the overall model complexity.

Correct Answer: A

Question 18

In Support Vector Machines (SVM), what does the "Margin" represent, and why do we want to maximize it?

- A. The total count of input features that are actively utilized by the Support Vector Machine model.
- B. The distance between the decision boundary and the closest data points (support vectors).
- C. The aggregate number of data points that are correctly classified within the training dataset.
- D. The total computational time required to fit the model to the training data until convergence.

Correct Answer: B

Question 19

How does the Kernel Trick help SVM handle non-linearly separable data?

- A. It removes the inner class data points to simplify the geometry of the decision boundary.
- B. It switches the underlying architecture from hyperplanes to decision trees for better classification.
- C. It implicitly maps data to higher dimensions where it becomes linearly separable.
- D. It modifies the optimization objective by changing the loss function from hinge loss to cross-entropy.

Correct Answer: C

Question 20

What is the effect of using a very large value of C in Soft Margin SVM?

- A. It mathematically simplifies the SVM, effectively converting it into a standard Linear Regression model.
- B. It forces all the model weights to approach zero, which serves as a strong mechanism to prevent overfitting.
- C. It becomes very tolerant of misclassifications, creating a wide margin that may lead to underfitting.
- D. It heavily penalizes misclassifications, leading to a narrow margin and potential overfitting.

Correct Answer: D

Question 21

How do Decision Trees learn to split data, and what method do they use (instead of Gradient Descent)?

- A. They use greedy search to find splits that maximize information gain (minimize impurity).
- B. They employ evolutionary genetic algorithms to randomly mutate and evolve the tree structure.
- C. They calculate the gradients of the loss function at every node to mathematically optimize splits.
- D. They utilize complex matrix inversion operations to algebraically determine the optimal thresholds.

Correct Answer: A

Question 22

What is the major disadvantage of using a single Decision Tree for predictions?

- A. It is considered a "black box" model and is extremely difficult to interpret compared to linear models.
- B. It has high variance (very unstable) — small changes in data can create completely different trees.
- C. It is fundamentally incapable of processing categorical data without complex pre-processing steps.
- D. It requires the input data to undergo extensive feature scaling and normalization before training.

Correct Answer: B

Question 23

How does Random Forest reduce the variance problem of single Decision Trees?

- A. It deliberately restricts the depth of every tree in the forest to be a shallow stump (depth of 1).
- B. It trains multiple trees on the exact same dataset and then selects the single best performing tree.
- C. It uses Bagging (Bootstrap Aggregating) to train multiple uncorrelated trees and averages their predictions.
- D. It employs Gradient Descent to iteratively optimize the structure and weights of every individual tree.

Correct Answer: C

Question 24

What is the fundamental difference in how Random Forest and Gradient Boosting train their trees?

- A. Random Forest trains trees in parallel (independent), while Gradient Boosting trains them sequentially.
- B. Random Forest utilizes linear models as base learners, whereas Gradient Boosting utilizes decision trees.
- C. Random Forest is theoretically guaranteed to overfit, while Gradient Boosting is immune to overfitting.
- D. Random Forest requires all features to be scaled, whereas Gradient Boosting handles unscaled data.

Correct Answer: A

Question 25

In Gradient Boosting, what does each new tree actually predict?

- A. The probability estimates for each class calculated directly from the original input features.
- B. The optimal coefficients and weights that would minimize the error for a linear regression model.
- C. The residual errors (mistakes) from the previous model's predictions.
- D. The original target values y from scratch, ignoring previous model predictions.

Correct Answer: C

Question 26

Which metrics are commonly used to measure node impurity when building Decision Trees?

- A. The Pearson Correlation Coefficient to measure linear relationships.
- B. Mean Squared Error (MSE) is the exclusive metric used for all tree types.
- C. Geometric distance metrics such as Euclidean Distance and Manhattan Distance.
- D. Gini Impurity or Entropy (Information Gain).

Correct Answer: D

Question 27

According to the "No Free Lunch" Theorem, what can we conclude about machine learning algorithms?

- A. No single algorithm performs best across all possible problems; we must experiment.
- B. Deep Learning neural networks will invariably outperform traditional Machine Learning models.
- C. Support Vector Machines (SVM) are theoretically proven to be the superior algorithm for any dataset.
- D. XGBoost is mathematically guaranteed to provide the optimal solution for all tabular data problems.

Correct Answer: A

Question 28

Why don't Tree-based models (like Decision Trees and Random Forests) require feature scaling?

- A. They essentially function by converting all input features into integer values during processing.
- B. They make threshold-based decisions (splits), which are invariant to feature scales.
- C. They utilize a specialized form of gradient descent that is mathematically invariant to feature scale.
- D. They completely disregard the actual values of the features and only look at their rank ordering.

Correct Answer: B

Question 29

What is the main risk of setting the Max Depth hyperparameter too high in Decision Trees?

- A. The computational speed of making predictions will increase significantly due to the depth.
- B. The tree will fail to capture complex patterns, resulting in a model that is too simple (underfitting).
- C. The tree will overfit by memorizing noise and creating overly complex boundaries.
- D. The tree will trigger an early stopping mechanism that prevents it from learning the data fully.

Correct Answer: C

Question 30

When working with a small dataset, what issue makes K-Fold Cross-Validation better than a simple train-test split?

- A. K-Fold Cross-Validation completely eliminates the necessity of having a separate validation set.
- B. K-Fold Cross-Validation has built-in capabilities to automatically perform feature engineering.
- C. K-Fold Cross-Validation is computationally more efficient and always runs faster than a simple split.
- D. A single random split can be "lucky" or "unlucky"; K-Fold tests on all data across multiple folds.

Correct Answer: D

Question 31

You are processing a dataset and encounter a column with missing values (NaN). Under what condition is "Imputation" (filling with Mean/Median) generally preferred over simply dropping the rows?

- A. When the dataset is massive and dropping a few rows won't affect the distribution.
- B. When the dataset is small or the missing rows contain valuable information in other columns.
- C. When the missing values are in the target column (labels) rather than the features.
- D. When the missing values are caused by a "Not Applicable" category (e.g., "Spouse Name" for a single person).

Correct Answer: B

Question 32

Which of the following scenarios best justifies using One-Hot Encoding instead of Label Encoding?

- A. The feature is ordinal with a clear hierarchy (e.g., "Low", "Medium", "High").
- B. The feature is a binary flag (e.g., "Yes", "No") that is already numeric.
- C. The feature is nominal with no intrinsic order (e.g., "Red", "Blue", "Green"), to prevent the model from assuming a mathematical relationship.
- D. The feature has high cardinality (e.g., "Zip Code" with 10,000 unique values), to reduce dimensionality.

Correct Answer: C

Question 33

What is a primary advantage of Linear Regression models?

- A. They automatically capture complex non-linear relationships without feature engineering.
- B. They are highly interpretable because the learned weights (w) directly indicate feature importance.
- C. They are immune to outliers and noise in the training data.
- D. They require no data preprocessing or feature scaling to function correctly.

Correct Answer: B

Question 34

What is a major disadvantage of Linear Regression?

- A. It has high variance and tends to overfit large datasets easily.
- B. It is a "Black Box" model that is difficult to explain to stakeholders.
- C. It often suffers from High Bias (Underfitting) because it cannot model non-linear data structures.
- D. It is computationally expensive to train compared to Neural Networks.

Correct Answer: C

Question 35

What is a unique advantage of k-Nearest Neighbors (k-NN) regarding its training phase?

- A. It has a "zero" training time complexity ($O(1)$) because it simply stores the data without learning parameters.
- B. It automatically selects the most relevant features during the training process.
- C. It creates a compact mathematical formula that discards the original training data to save memory.
- D. It trains significantly faster than Linear Regression on massive datasets.

Correct Answer: A

Question 36

What is a significant disadvantage of k-Nearest Neighbors (k-NN) during prediction (inference)?

- A. It requires a differentiable loss function, making it incompatible with certain metrics.
- B. It is prone to underfitting because it creates linear decision boundaries.
- C. It is computationally slow ($O(N)$) because it must calculate distances to all training points for every new prediction.
- D. It cannot handle multi-class classification problems effectively.

Correct Answer: C

Question 37

What is a key advantage of Support Vector Machines (SVM)?

- A. They are extremely fast to train on datasets with millions of samples.
- B. They work exceptionally well in high-dimensional spaces (even when features $>$ samples).
- C. They are insensitive to noise and outliers regardless of the hyperparameters used.
- D. They provide direct probability estimates natively without extra computation.

Correct Answer: B

Question 38

What is a notable disadvantage of Support Vector Machines (SVM)?

- A. They cannot model non-linear relationships even with Kernels.
- B. They generally perform poorly on small datasets.
- C. They are sensitive to feature scaling and require data normalization to function properly.
- D. They are "Lazy Learners" that do not generalize well to unseen data.

Correct Answer: C

Question 39

What is a major advantage of Decision Trees related to data preprocessing?

- A. They can automatically impute missing values using advanced regression techniques.
- B. They require no feature scaling or normalization because they use rule-based splits.
- C. They automatically dimensionality reduce the data using PCA before training.
- D. They convert all unstructured text data into numerical vectors automatically.

Correct Answer: B

Question 40

What is the primary disadvantage of a single Decision Tree?

- A. It has High Bias and struggles to learn even simple patterns from the training data.
- B. It is a "Black Box" model that is impossible for humans to interpret.
- C. It has High Variance (Instability), where small changes in data lead to completely different trees.
- D. It requires expensive GPU resources to train effectively.

Correct Answer: C

Question 41

How does Random Forest improve upon Decision Trees?

- A. It trains a single tree to maximum depth to ensure zero training error.
- B. It is robust to overfitting and reduces variance by averaging the predictions of many uncorrelated trees.
- C. It uses Gradient Descent to mathematically optimize the split thresholds for higher precision.
- D. It converts the classification problem into a regression problem for smoother boundaries.

Correct Answer: B

Question 42

What is a disadvantage of Random Forest compared to a single Decision Tree?

- A. It is much more likely to overfit the data than a single tree.
- B. It requires extensive hyperparameter tuning to get any reasonable results.
- C. It has slower inference (prediction) speed because data must pass through hundreds of trees.
- D. It loses the ability to handle categorical data directly.

Correct Answer: C

Question 43

What is a key advantage of Gradient Boosting (e.g., XGBoost)?

- A. It is the fastest algorithm to train because all trees are built in parallel.
- B. It provides State-of-the-Art (SOTA) performance on tabular data and handles missing values natively.
- C. It is a linear model that guarantees global convergence to the optimal solution.
- D. It requires absolutely no hyperparameter tuning to achieve top performance.

Correct Answer: B

Question 44

What is a disadvantage of Gradient Boosting?

- A. It is extremely sensitive to hyperparameters (e.g., Learning Rate, Depth) and easy to overfit if not tuned.
- B. It cannot handle large datasets because it loads all data into the CPU cache.
- C. It is restricted to binary classification tasks and cannot perform regression.
- D. It produces high-bias models that systematically underfit complex data.

Correct Answer: A

Question 45

What is the fundamental difference between Deep Learning and traditional Machine Learning regarding feature extraction?

- A. Deep Learning requires manual feature engineering, while Machine Learning automates it.
- B. Deep Learning automates feature extraction through layers, while Machine Learning often relies on manual engineering.
- C. Deep Learning can only handle tabular data, while Machine Learning handles images and text better.
- D. Deep Learning uses CPU exclusively, while Machine Learning requires GPU acceleration.

Correct Answer: B

Question 46

In a neural network, what is the primary role of the Activation Function?

- A. To initialize the weights to random small values before training begins.
- B. To introduce non-linearity, allowing the network to learn complex patterns.
- C. To calculate the gradient of the loss function during backpropagation.
- D. To normalize the input data to a standard range of 0 to 1.

Correct Answer: B

Question 47

You have a fully connected layer with 4 input neurons and 3 output neurons. How many learnable parameters does this layer have?

- A. 12 parameters
- B. 7 parameters
- C. 15 parameters
- D. 16 parameters

Correct Answer: C

Question 48

You have a Neural Network with the following architecture:

- **Input Layer:** 4 neurons
- **Hidden Layer 1:** 5 neurons
- **Hidden Layer 2:** 4 neurons
- **Hidden Layer 3:** 3 neurons
- **Output Layer:** 3 neurons

What is the **total** number of learnable parameters in the entire network?

- A. 56 parameters
- B. 65 parameters
- C. 76 parameters
- D. 82 parameters

Correct Answer: C

Question 49

What is the "Vanishing Gradient" problem, and which activation function helps mitigate it?

- A. Gradients become too large, causing instability; Sigmoid fixes this.
- B. Gradients approach zero, stopping learning; ReLU helps prevent this.
- C. The loss function becomes negative; Tanh corrects the range.
- D. The model overfits immediately; Linear activation prevents it.

Correct Answer: B

Question 50

What is the main purpose of the Softmax activation function in the output layer?

- A. To convert raw logits into a probability distribution that sums to 1.
- B. To set all negative values to zero to induce sparsity.
- C. To normalize the weights of the network to prevent overfitting.
- D. To speed up the calculation of the derivative during backpropagation.

Correct Answer: A

Question 51

In the context of optimization, what does Momentum help SGD (Stochastic Gradient Descent) achieve?

- A. It forces the learning rate to decrease to zero immediately.
- B. It helps the optimizer navigate through ravines and dampens oscillations.
- C. It removes the need for backpropagation by guessing the gradient.
- D. It ensures that the loss function is always convex.

Correct Answer: B

Question 52

Why is Unsupervised Learning often described as finding "hidden structures" in data?

- A. Because it uses labeled data to predict hidden target variables.
- B. Because it hides the input features to test the model's robustness.
- C. Because There is no prior knowledge about the target.
- D. Because it requires a hidden layer in a neural network to function.

Correct Answer: C

Question 53

How does the K-Means clustering algorithm determine which cluster a data point belongs to?

- A. By calculating the probability gradient of the density function.
- B. By using a decision tree to split the data based on entropy.
- C. By maximizing the margin between different data groups.
- D. By assigning it to the cluster with the nearest centroid (mean).

Correct Answer: D

Question 54

What is a major limitation of K-Means clustering?

- A. It cannot handle numerical data and requires categorical inputs.
- B. You must specify the number of clusters (k) manually beforehand.
- C. It is computationally too expensive to run on datasets larger than 100 rows.
- D. It always produces the exact same result regardless of initialization.

Correct Answer: B

Question 55

What is the primary goal of Principal Component Analysis (PCA)?

- A. To increase the dimensionality of the data to find more complex patterns.
- B. To classify data points into distinct categories based on labels.
- C. To reduce dimensionality while preserving the maximum variance (information).
- D. To remove all noise from the data by setting small values to zero.

Correct Answer: C

Question 56

When using t-SNE for visualization, what key characteristic distinguishes it from PCA?

- A. t-SNE is a linear technique, whereas PCA is non-linear.
- B. t-SNE is better at preserving local structures (clusters) in non-linear data.
- C. t-SNE is significantly faster to compute than PCA on large datasets.
- D. t-SNE can only reduce data to exactly one dimension.

Correct Answer: B

Question 57

What is an Autoencoder primarily designed to learn?

- A. A mapping from input features to discrete class labels.
- B. A function that compresses (encodes) and then reconstructs (decodes) the input.
- C. A prediction of future time-series values based on past data.
- D. A policy for an agent to maximize rewards in an environment.

Correct Answer: B

Question 58

How can an Autoencoder be used for Anomaly Detection?

- A. By classifying anomalies as "Class 1" and normal data as "Class 0".
- B. By automatically deleting anomalous rows during the training process.
- C. By using the encoder part to visualize the data in 2D space.
- D. By training on normal data; anomalies will have a high reconstruction error (loss).

Correct Answer: D

Question 59

In the context of PCA, what does the first Principal Component (PC1) represent?

- A. The direction in the data that captures the least amount of noise.
- B. The direction along which the data varies the most (maximum variance).
- C. The feature that is most correlated with the target variable.
- D. The average vector of all the data points in the dataset.

Correct Answer: B

Question 60

What is the "bottleneck" in an Autoencoder architecture?

- A. The input layer, which restricts the amount of data the model can process.
- B. The output layer, which limits the resolution of the reconstructed image.
- C. The compressed middle layer (Latent Space) that forces the network to learn efficient representations.
- D. The activation function, which saturates and prevents gradients from flowing.

Correct Answer: C

Question 61

The Adam optimizer is widely used because it effectively combines two key mechanisms from previous algorithms. What are they?

- A. **L1 and L2 Regularization** to prevent overfitting during weight updates.
- B. **Momentum** (to accelerate in the correct direction) and **RMSProp** (adaptive learning rates for each parameter).
- C. **Grid Search and Random Search** to find the optimal path to the global minimum.
- D. **Backpropagation and Forward Propagation** to calculate gradients simultaneously.

Correct Answer: B

Question 62

Why is Forward Propagation necessary before Backpropagation can occur?

- A. To calculate the current loss (error) so we know "how wrong" the model is.
- B. To update the weights using the previous batch's gradients.
- C. To randomly initialize the weights for the current epoch.
- D. To determine which activation function should be used for the layer.

Correct Answer: A

Question 63

What is the main advantage of using Mini-Batch Gradient Descent over processing the entire dataset at once (Batch GD)?

- A. It guarantees convergence to the global minimum for non-convex functions.
- B. It eliminates the need for a learning rate hyperparameter.
- C. It calculates the exact gradient, ensuring a smooth descent path.
- D. It fits in memory and provides faster weight updates with some beneficial noise.

Correct Answer: D

Question 64

You have a **very big dataset** and you want to group customers into market segments but you do not have any labels for "Segment A" or "Segment B," which approach should you use?

- A. PCA
- B. T-SNE
- C. Auto-Encoders
- D. K-means.

Correct Answer: D

Question 65

You are analyzing a dataset with a complex, curved structure (like a "Swiss Roll" or a coiled spring). You observe that **PCA** completely fails to unfold the shape, smashing different layers together, while **t-SNE** successfully separates the curve into a flat band. **What is the mathematical reason for this difference?**

- A. PCA is a **Linear** algorithm that projects data onto a flat plane, making it incapable of understanding curved "Manifolds."
- B. t-SNE uses a "Kernel Trick" (similar to SVM) to project the data into a higher infinite-dimensional space where it becomes linear.
- C. PCA requires the data to be labeled (Supervised), while t-SNE uses the labels to guide the unfolding process.
- D. t-SNE rotates the data axes to find the maximum variance, while PCA relies on probabilistic neighbor selection.

Correct Answer: A